

Good Gradings of Generalized Incidence Rings

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October 11, 2018

Abstract

This inquiry is based on both the construction of generalized incidence rings due to Gene Abrams and the construction of good group gradings of incidence algebras due to Molli Jones. We provide conditions for a generalized incidence ring to be graded isomorphic to a subring of an incidence ring over a preorder. We also extend Jones's construction to good group gradings for incidence algebras over preorders with crosscuts of length one or two.

1 Overview

Unless otherwise stated we use multiplicative notation for all semigroup, monoid, or group operations and the identity is denoted by 1.

Suppose G is a semigroup and S is a ring which does not necessarily contain a multiplicative identity. We say S is a G -graded ring if there is a direct sum